CP020003

ARTIFICIAL INTELLIGENCE



Khon Kaen University · 2026

Final Week

Final Exam Guide

What to expect, how to prepare, and how to earn the points — Weeks 9 to 12

Instructor: Teerapong Panboonyuen (P'Kao)

https://github.com/kaopanboonyuen/CP020003_ArtificialIntelligence_2026s1

The exam at a glance

3 h

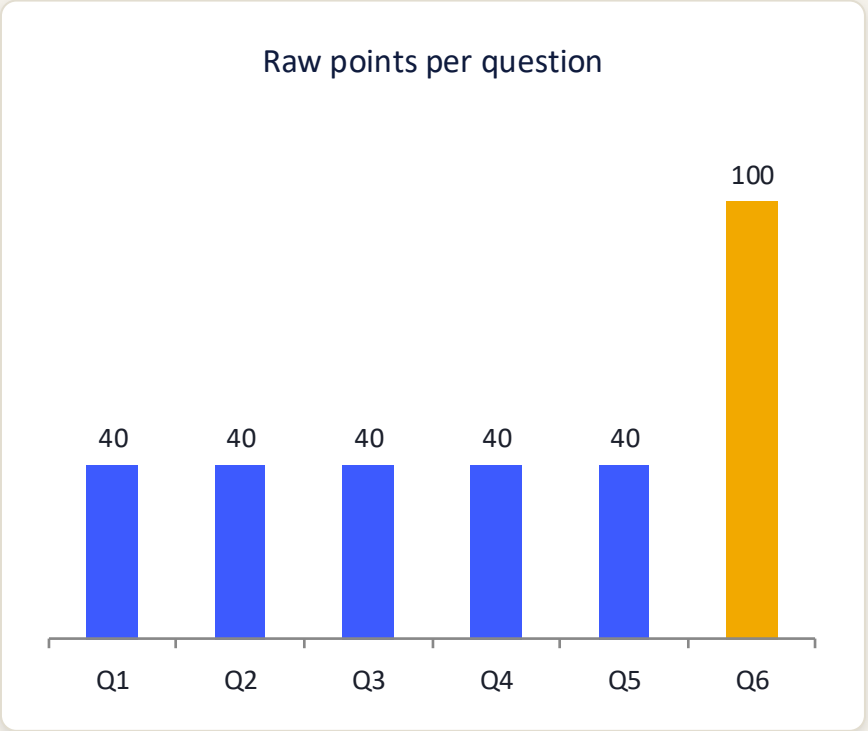
Full 180 minutes — plan to use all of it

6

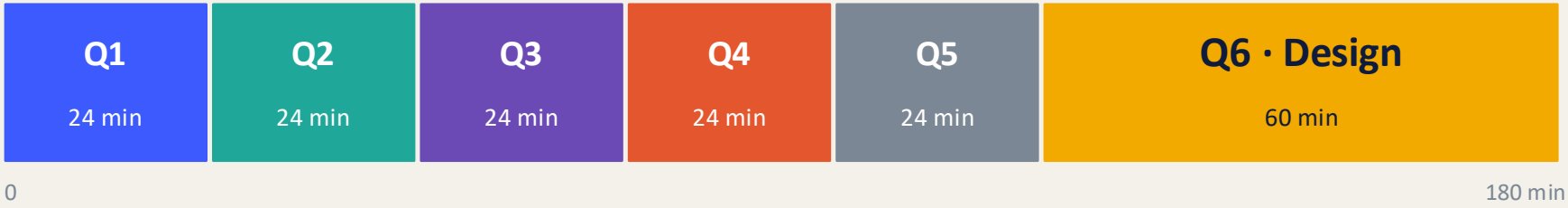
Big questions, each with sub-parts

300 → 30%

Raw points ÷ 10 = your final 30%



A suggested pace for 180 minutes



Skim first

Spend the first minutes reading Q6 so your brain designs in the background while you collect quick points.

Bank the quick points

Q1–Q5 are built for fast, clear marking. Do not let one stuck item eat your Q6 time.

Keep a review buffer

Leave a few minutes to re-check calculations and fill any blank answer boxes.

Q1-Q5: quick, clearly marked items

Q1 to Q4 follow Weeks 9 to 12 in order. Q5 mixes everything. Each question uses a similar mix of formats:



True / False

A statement to judge. Watch for words like “always” and “only”.



Multiple choice

One best answer. Look for the option that matches the course’s reasoning.



Fill in the blank

Short key terms, numbers, or model names from the lectures.



Matching

Link tasks, metrics, or scenarios to the right concept.



Short answer & calculation

Explain briefly or compute by hand. Show your steps.

Q1 · Time series (Week 9)



WHAT THE QUESTION TESTS

Be ready to EXPLAIN

- Why time series differ from ordinary tabular data
- Stationarity, differencing, and what the ADF test decides
- What p, d, q mean; what ACF and PACF are used for
- Why a naive baseline matters
- How classical, tabular-ML, and deep models relate

Be ready to DO

- Compute MAE, RMSE, and MAPE by hand from a small table
- Choose a metric for a situation and justify it
- Spot an invalid train/test setup
- Read the model ladder: AR/MA → ARIMA → lags + trees → LSTM → attention

Expect: true/false, multiple choice, key-term blanks, and one hand calculation.

Q2 · Banking regression (Week 10)



WHAT THE QUESTION TESTS

Be ready to EXPLAIN

- Regression versus classification
- Cleaning real data and the idea of data leakage
- Linear vs Ridge / Lasso / ElasticNet
- Random forest vs gradient boosting
- Why lenders need explainable decisions (SHAP, ECOA)

Be ready to DO

- Apply the “would I know this before quoting?” test to a feature
- Interpret the sign of a coefficient
- Read a model leaderboard and R^2
- Explain how a GenAI layer should (and should not) be used in lending

Expect: blanks on model facts, conceptual true/false, and short reasoning answers.

Q3 · Computer vision (Week 11)

WHAT THE QUESTION TESTS

Be ready to EXPLAIN

- Six tasks: classification, detection, OBB, segmentation, depth, pose
- What each task outputs and when to choose it
- Label formats (what each number in a label line means)
- Metrics: Top-1/Top-5, IoU, mAP50, mAP50-95, AbsRel, δ1, OKS

Be ready to DO

- Match a real-world need to the right task
- Compute IoU for two boxes from their coordinates
- Decide if a detection counts at a given threshold
- Tell which metrics are “higher is better”

Expect: matching, multiple choice on label formats, and one geometry calculation.

Q4 · Generative AI (Week 12)

WHAT THE QUESTION TESTS

Be ready to EXPLAIN

- Autoregressive LLMs versus diffusion models
- What num_inference_steps and guidance_scale do
- CLIP score, BLIP captioning, and the round-trip game
- The two checks for a good inpaint
- Distillation, flow matching, DiT, and the 2026 trend

Be ready to DO

- Compare two model families on method and training signal
- Predict the effect of a knob being too low or too high
- Explain why a metric needs a partner metric
- Describe what survives and what drifts across modalities

Expect: true/false, multiple choice, blanks on key terms, and short comparison answers.

Q5 · Connecting the whole course

This one mixes Weeks 9–12. Think like a practitioner choosing and checking tools.

Pick the tool

Given a short business scenario, which technique fits: forecasting, regression, detection, captioning, inpainting…?

Match the metric

Which metric belongs to which kind of problem, and what does it tell you?

Find the flaw

A short description of a flawed pipeline. Name the mistake and how to fix it.

Think responsibly

Why separate decision-making from explanation? Name risks such as bias, privacy, and fake content.

QUESTION 6 · 100 RAW POINTS

Design an AI system

A client describes what they want. You decide how to build it, and why.

What the design question looks like

You will be given

- A business scenario with several needs
- Data that is available (and messy)
- Real constraints: budget, hardware, laws, deadline

Several needs, possibly from different weeks of the course.

1. **Problem formulation**
2. **Data & preparation**
3. **Model choice + baseline**
4. **Evaluation & success criteria**
5. **System architecture**
6. **Risks, ethics, monitoring**
7. **Trade-offs & priorities**

How design answers earn points

Tie choices to constraints

No GPU at the edge? Deadline? Privacy law? Say how each shaped your decision.

Always name a baseline

A simple reference tells the client whether the fancy model is worth it.

Make success measurable

Pair a metric with a target, for example “better than baseline by a stated margin”.

Show the data flow

Sources → preparation → models → serving → users → feedback. Say what runs where.

Anticipate what can go wrong

Leakage, drift, privacy, bias, wrong or fake outputs, who can override.

Justify, don’t just name

“We use X because Y, compared against Z” beats a list of buzzwords.

Rules of thumb from the course

- **Never shuffle. Never peek at the future.**
- **Beat the baseline before claiming success.**
- **If a feature cheats, the model lies.**
- **Compare error metrics: the gap tells a story.**
- **Explainability is required in regulated work.**
- **Every translation between modalities is lossy.**

Formulas worth knowing cold

Calculations appear in the exam. A calculator is helpful, so practise the steps.

MAE

average of |actual − predicted|

RMSE

square root of the average squared error

MAPE

average of |error| ÷ |actual|, as a %

IoU

overlap area ÷ combined area

R^2

share of variance explained (0 = no better than the mean)

Top-1 / Top-5

correct label is the best guess / among the best five

A simple study plan

1	**Re-read each lecture**	One page of notes per week: key terms, metrics, and “when to use what”.
2	**Re-run the notebooks**	Look at the outputs and ask why each result came out that way.
3	**Practise by hand**	Do small calculations and write short explanations without notes.
4	**Design out loud**	Invent a client scenario and talk through all seven design steps.

Good luck!

Bring a pen and a calculator. Read each question twice. Show your reasoning.